

Task 16 — Enterprise Multi-Tenancy & RBAC

Data Analyst · Sprint D - Enterprise Readiness · Task Brief

DATA ANALYST

PlaceMux · Phase 3 · Task Brief

TASK 16 / 25

Enterprise Grade · Expert

Sprint D - Enterprise Readiness · Mastery 85%

Focus

Build multi-tenant analytics: per-tenant reporting with strict isolation and no cross-tenant leakage.

What to build / complete

- Per-tenant analytics and reporting with enforced isolation
- Aggregate benchmarks that cannot be reverse-engineered to a single tenant
- Verified no cross-tenant data leakage in any report

Execution — work through in order

Phase 3 gives you a goal and a bar, not a recipe. These stages are the discipline; the design decisions are yours to make and defend.

Stage A — Understand & set the bar

1. Re-read the Expected Output and the scoring parameters so you build to the bar, not past it.
2. Confirm every prerequisite and upstream input is actually in your hands, not merely promised.
3. Restate, in one sentence, what 'good' means here: Each enterprise sees rich analytics about themselves and learns nothing about anyone else.

Stage B — Build: Per-tenant analytics and reporting with enforced isolation

1. Define the metric, its source & the decision

Define precisely what “Per-tenant analytics and reporting with enforced isolation” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage C — Build: Aggregate benchmarks that cannot be reverse-engineered to a single tenant

1. Define the metric, its source & the decision

Define precisely what “Aggregate benchmarks that cannot be reverse-engineered to a single tenant” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage D — Build: Verified no cross-tenant data leakage in any report

1. Define the metric, its source & the decision

Define precisely what “Verified no cross-tenant data leakage in any report” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage E — Integrate, verify & make demoable

1. Wire your work into the end-to-end flow and run the whole journey once, start to finish.
2. Show a tenant report and prove no other tenant's data is inferable from it.
3. Break it on purpose: force the failure path and confirm it degrades the way you designed.
4. Prepare a live demo on real data. Not slides, not screenshots, not 'it works on my machine'.

Done when (demoable by end of task)

- “Per-tenant analytics and reporting with enforced isolation” is complete, real, and demoable end-to-end.
- “Aggregate benchmarks that cannot be reverse-engineered to a single tenant” is complete, real, and demoable end-to-end.
- “Verified no cross-tenant data leakage in any report” is complete, real, and demoable end-to-end.
- Good looks like: Each enterprise sees rich analytics about themselves and learns nothing about anyone else.

Worry if any of these are true

- Benchmarks computed over too few tenants.
- Warehouse without tenant isolation.
- Calling an experiment early because it looked good on day two.

Hand-off

- Tenant reporting to Backend/Frontend. Make sure the next person can pick it up without you.